Cross-Modal Commutator Rotation for Multimodal Sarcasm Detection: Relative Attention Geometry and Controlled Evaluation

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***Abstract*—Multimodal sarcasm detection requires understanding how visual context qualifies textual meaning, rather than simply maximizing image–text alignment. We propose Cross-Modal Commutator Rotation (CMCR), a geometrically constrained mechanism for adapting cross-modal attention. CMCR constructs sample-conditioned low-rank rotations from first- and second-order modality statistics and applies distinct coordinate frames to visual and non-visual queries and keys. For fixed layer inputs, this design preserves same-frame query–key inner products in exact arithmetic while allowing cross-frame attention scores to change. Experiments on MMSD2.0 show that CMCR with Qwen3-VL-8B-Thinking achieves 87.21% accuracy and 87.12% macro-F1. Cross-scale comparisons, ablation studies, and diagnostic analyses characterize the effects of its components and the resulting predictive behavior. CMCR provides a geometrically explicit approach to studying and adapting modality-relative interactions for multimodal sarcasm detection. The source code is publicly available at https://github.com/CICC-XJTLU/CMCR.**

***Index Terms*—Multimodal sarcasm detection, attention geometry, covariance commutator, orthogonal adaptation, vision–language models.**

I. Introduction

RELIABLE sarcasm recognition requires interpreting a caption's evaluative stance in relation to the situation depicted alongside it. Praise can function as criticism when the image supplies evidence that the stated evaluation is difficult to sustain. The relevant cue is therefore not a positive word or an unpleasant scene in isolation, but how the referenced visual content qualifies the textual appraisal. This distinction places the burden on cross-modal reasoning rather than sentiment cues or superficial correspondence. The annotation and shortcut concerns documented in MMSD2.0 further motivate evaluating whether a detector uses that relationship rather than dataset-specific regularities [1].

The problem is not simply to maximize image–text alignment. A sincere complaint can be fully consistent with a negative image, whereas a sarcastic compliment can refer accurately to the same objects while reversing their evaluation. Conversely, an apparently incompatible image may be irrelevant rather than ironic. These possibilities suggest treating correspondence and evaluation as related but distinct aspects of inference. Increasing the total amount of visual information entering a representation does not specify how a text query should read that information.

Existing systems have progressed from hierarchical multimodal fusion to input-dependent interaction mechanisms, as summarized by recent survey work [2], [3]. Factual and affective incongruity need not coincide: separating them can provide a more precise account of why an image changes the interpretation of a sentence [4]. Contrastive pretraining learns broad image–text correspondences [5], [6], and lightweight learned interfaces can connect frozen visual and language components [7]. Large vision–language models can then use retrieved demonstrations or contrasting rationales to organize contextual information [8], [9]. Such approaches motivate an additional question inside the model: can adaptation change the relative geometry of cross-modal attention while constraining the direct disturbance to within-modality scores? We address this question through an operator inside native attention.

Attention is a natural intervention site because a query–key inner product determines how information is read before values are aggregated. Unconstrained transformations can change both within- and cross-modal scores, making their effects difficult to separate. Applying the same orthogonal transformation to every query and key has the opposite problem: it leaves their inner products unchanged and therefore cannot directly implement the desired relative change. Orthogonal fine-tuning constrains weight transformations geometrically [10], whereas representation fine-tuning intervenes on hidden activations [11]. We exploit a different distinction within attention: a shared coordinate transformation versus a modality-relative transformation. Visual positions use the inverse of the frame used by non-visual positions, with both frames constructed in a learned low-dimensional head subspace.

We introduce Cross-Modal Commutator Rotation (CMCR). In each selected attention layer, the model summarizes caption and visual hidden states through normalized means and trace-normalized covariances. The antisymmetric outer-product difference of the means describes an oriented first-order interaction. The covariance commutator describes whether the

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two second-order operators commute in the learned statistical space. A conditional controller combines these terms with a learned static skew-symmetric component. The resulting bounded generator is mapped to an orthogonal rotation through the Cayley transform and applied to queries and keys after their native rotary positional encoding.

For a fixed layer input and a shared key–value head group, CMCR preserves same-frame query–key logits in exact arithmetic while allowing cross-frame logits to change. The controller derives these frames from the complete image–caption input for full-input classification.

We evaluate CMCR as a complete multimodal sarcasm classifier and examine its static-frame and no-commutator variants under a common training and validation protocol. The 8B model attains 87.21% accuracy and 87.12% macro-F1. Its accuracy exceeds the static-frame and no-commutator variants by 0.17 and 0.29 percentage points, respectively. Across the 4B and 8B backbones, the full formulation achieves consistent gains over the no-commutator variant. Cross-scale results, paired decisions, checkpoint interventions, confidence analysis, and resource measurements characterize the model from complementary predictive and geometric perspectives.

The contributions are threefold:

- We formulate a sample-conditioned relative attention operator using first- and second-order modality statistics, bounded skew-symmetric generators, and low-rank orthogonal frames within native multimodal attention.
- We establish the operator's conditional same-frame logit invariance and cross-frame perturbation bound, and implement it after native positional rotation with the original value path and attention mask retained.
- We provide a controlled evaluation across two backbone sizes, separating independently trained ablations from fixed-checkpoint interventions and quantifying uncertainty, numerical-control effects, classwise errors, calibration, and computational cost.

II. Related Work

A. Multimodal sarcasm detection and target identification

Multimodal sarcasm detection has developed from feature fusion toward more explicit descriptions of incongruity. Hierarchical fusion integrates textual, visual, and image-attribute information [2]. Attention-based modeling separates contradiction within a sentence from disagreement across image and text [12]. Knowledge-enhanced hierarchical modeling further distinguishes atomic from compositional congruity [13]. These designs expose different interaction levels rather than assuming that a single pooled representation captures the relevant contrast. The broader literature covers a range of modalities and supervision settings [3].

Later detectors make the interaction more selective. DynRT activates different transformer routes for individual image–text pairs [14], while DIP separates factual and affective incongruity [4]. MMSD2.0 revisits annotation reliability and shortcut behavior, emphasizing the need to evaluate these mechanisms on a cleaner benchmark [1]. CMCR models these interactions through native queries and keys, using a covariance commutator to describe the relative geometry of second-order modality statistics.

Target identification asks which entity receives a sarcastic evaluation, rather than only whether sarcasm is present. CofiPara combines contrasting rationales with coarse-to-fine target identification [9]. Retrieval-assisted detection supplies visual instructions and demonstrations for a new post [8]. These systems organize contextual evidence through target-level or prompt-based interfaces, whereas CMCR adapts cross-modal attention directly through hidden-state statistics.

B. Visual grounding and intermediate semantic representations

Large-scale pretraining provides several ways to connect visual and linguistic information. CLIP learns transferable image–text representations through contrastive supervision [5]; ALBEF aligns the modalities before cross-modal fusion and uses momentum distillation [6]. A broader taxonomy distinguishes representation, alignment, and fusion as separate challenges in multimodal learning [15]. These approaches establish useful correspondences, but global correspondence is not equivalent to pragmatic compatibility. A sarcastic caption may correctly name the visible objects while expressing an evaluation that the scene undermines.

Different interfaces determine how those correspondences reach a language model. BLIP-2 uses a Querying Transformer between frozen visual and language components [7]. Transformer-based multimodal learning uses attention mechanisms for alignment and fusion [16], while InstructBLIP makes the visual extraction instruction-aware [17]. For explicit localization, position-guided text prompts incorporate object–location associations into vision–language pretraining [18]. CMCR retains the pretrained interface: visual and caption masks select the states for moment computation, and a learned basis defines the transformation in attention-head space.

Parameter-efficient adaptation changes a pretrained model through a restricted trainable interface. LoRA learns low-rank weight updates [19]; DoRA separates weight magnitude and direction [20]; VeRA shares fixed low-rank matrices and learns scaling vectors [21]. ReFT instead learns interventions on hidden representations [11]. Orthogonal fine-tuning and butterfly-factorized adaptation constrain transformations geometrically [10], [22]. CMCR constructs sample-conditioned activation frames acting directly on queries and keys, and trains this geometric pathway jointly with the classification head and language-side low-rank updates. Independently trained static and no-commutator controls examine input dependence and the explicit second-order term within the geometric formulation.

Native attention structure also constrains where such an operator can be applied. RoFormer introduces rotary positional structure into query–key interactions [23], while grouped-query attention shares key–value heads across several query heads [24]. CMCR applies its frames after native positional rotation and shares each frame within the corresponding KV group, establishing the placement and sharing conditions for fixed-input same-frame invariance.

C. Visual faithfulness and feedback-based alignment

Visual reliability needs tests beyond aggregate answer accuracy. POPE examines object hallucination through controlled object-presence questions [25]; MM-Vet evaluates combinations of visual–language capabilities rather than a single undifferentiated ability [26]. These designs motivate decomposed evaluation of visual–language behavior. Hallucination mitigation can intervene during generation. Visual contrastive decoding compares output distributions under original and distorted visual inputs [27]. OPERA uses an attention-based over-trust penalty and retrospective token-selection adjustments [28].

Our classwise, paired-decision, and confidence analyses examine complementary aspects of the trained predictor. CMCR acts inside attention and returns a class distribution in one pass. Its common-frame control removes the direct modality-relative score effect in exact arithmetic, while comparison with bypass provides a finite-precision reference for checkpoint interventions.

Preference learning offers another alignment route. Direct Preference Optimization learns from response preferences without a separately trained reward model or an online reinforcement-learning loop [29]. CMCR initializes the backbone from the pretrained Thinking checkpoint and optimizes detection labels with a rotation-energy penalty.

Work on attention interpretability motivates evaluating computational behavior alongside predictive performance [30]. We therefore examine CMCR through three complementary views: operator geometry, classification performance, and sensitivity to fixed-checkpoint interventions.

III. Proposed Method

A. Sample-Wise Moment Geometry inside Native Attention

Given an image I, its caption T, and label $y \in \{0,1\}$, the native processor constructs the multimodal input. The visual positions $\mathcal{V}$ are identified by the image-token identifier; caption positions $\mathcal{T}$ lie after the visual terminator and before the user-message terminator. Template positions are excluded from moment statistics. Figure 1 shows the classification pathway and the three insertion layers, with one-based indices $\{12, 24, 32\}$.

Let $H^\ell \in \mathbb{R}^{S\times d}$ be the hidden tensor received by self-attention at a selected layer. The 8B backbone has $d = 4096$, 36 decoder layers, 32 query heads, eight key–value groups, and head dimension $d_h = 128$. A layer-specific normalization and a shared downward-projection construct a rank-r statistical space:

$$Z^\ell = \tanh\big(\mathrm{LN}_\ell(H^\ell)W_z\big), \qquad W_z \in \mathbb{R}^{d\times r}, \quad r = 16. \tag{1}$$

The shared projection is applied independently at each insertion layer. Each layer computes sample-wise statistics from its own current input.

For $m \in \{t, v\}$, let $\mathcal{M}_t = \mathcal{T}$, $\mathcal{M}_v = \mathcal{V}$, and $n_m = |\mathcal{M}_m|$. With $\epsilon = 10^{-6}$, the implemented statistics are

$$\begin{aligned} \bar{z}_m &= n_m^{-1} \sum_{i\in\mathcal{M}_m} z_i, \quad \mu_m = \bar{z}_m / \max(\|\bar{z}_m\|_2, \epsilon), \\ \widetilde{C}_m &= n_m^{-1} \sum_{i\in\mathcal{M}_m} (z_i - \bar{z}_m)(z_i - \bar{z}_m)^\top, \\ C_m &= \widetilde{C}_m / \max(\operatorname{tr}\widetilde{C}_m, \epsilon). \end{aligned} \tag{2}$$

The covariance is centered on the unnormalized mean and divided by n_m. The moments helper guards empty masks with a count floor of one; the input collator rejects an empty caption or visual mask. Mean directions discard magnitude, whereas trace-normalized covariance captures the relative allocation of within-modality variation. Both normalizations use explicit floors to keep degenerate inputs numerically defined.

Define $\mathcal{B}_\tau(K) = K/\max(\|K\|_F, \tau)$. The first- and second-order geometric terms are

$$\begin{aligned} K_\mu &= \mu_t\mu_v^\top - \mu_v\mu_t^\top, \quad \widehat{K}_\mu = \mathcal{B}_1(K_\mu), \\ K_\Sigma &= C_tC_v - C_vC_t, \quad \widehat{K}_\Sigma = \mathcal{B}_{0.02}(K_\Sigma). \end{aligned} \tag{3}$$

Both matrices are skew-symmetric. The first term vanishes when the mean directions are parallel or antiparallel; the second vanishes for commuting covariance operators, including covariances sharing an eigenbasis. Supervised learning incorporates these first- and second-order geometric signals into the classifier. Figure 2 separates the two constructions.

B. Bounded Commutator Generators and Orthogonal Frames

Each insertion layer has a controller producing two coefficients for each KV group. Its input concatenates six rank-r vectors:

$$\begin{aligned} s = [\mu_t; \mu_v; \mu_t \odot \mu_v; |\mu_t - \mu_v|; \\ \operatorname{diag}(C_t); \operatorname{diag}(C_v)], \\ (a_{\ell g}, b_{\ell g}) = [\sigma(f_\ell(s))]_g. \end{aligned} \tag{4}$$

Here f_ℓ is a $96 \to 64 \to 16$ GELU MLP. The 16 outputs represent two coefficients for each of eight groups. A learned matrix $P_{\ell g}$ supplies a static skew component. The bounded generator is

$$\begin{aligned} B_{\ell g} &= \tfrac{1}{2}(P_{\ell g} - P_{\ell g}^\top), \\ A_{\ell g} &= 0.35\sigma(\rho_{\ell g})\, \mathcal{B}_1\Big(B_{\ell g} + a_{\ell g}\widehat{K}_\mu + b_{\ell g}\widehat{K}_\Sigma\Big). \end{aligned} \tag{5}$$

Thus $A_{\ell g}^\top = -A_{\ell g}$ and $\|A_{\ell g}\|_F \le 0.35\sigma(\rho_{\ell g})$. The coefficient ρ is initialized to -1. The static component expresses a learned prior; the two moment terms provide sample conditioning.

A Cayley transform gives a rank-space rotation, and thin QR gives an orthonormal basis in the head space:

$$\begin{aligned} R_{\ell g} &= (I_r - A_{\ell g})^{-1}(I_r + A_{\ell g}), \\ U_{\ell g} &= \mathrm{qf}(U_{\ell g}^0), \qquad U_{\ell g}^0 \in \mathbb{R}^{d_h\times r}, \\ G_{\ell g} &= I_{d_h} + U_{\ell g}(R_{\ell g} - I_r)U_{\ell g}^\top. \end{aligned} \tag{6}$$

The code solves a linear system in FP32 rather than explicitly forming an inverse. For real skew-symmetric A, $I - A$ is nonsingular and the Cayley transform is orthogonal. Since $U^\top U = I_r$, G is orthogonal and acts as the identity on

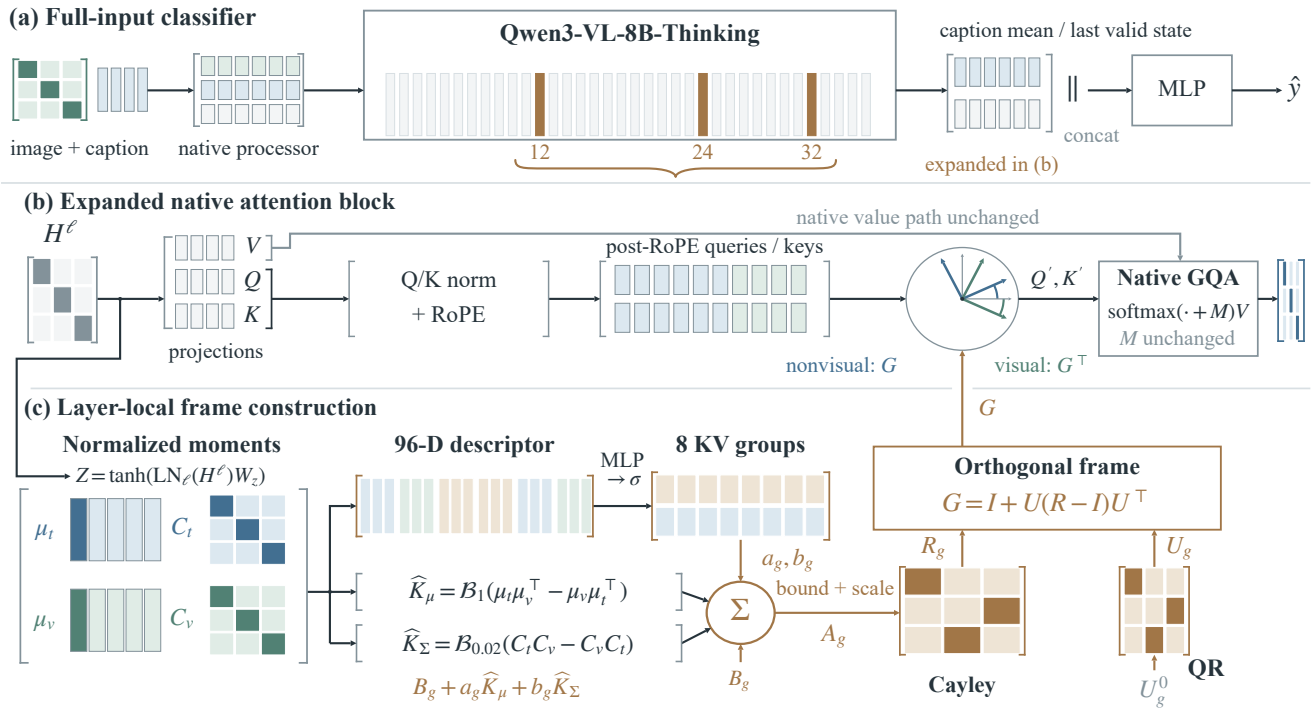


Fig. 1. CMCR inside native multimodal attention. Selected decoder layers derive sample-wise modality moments, construct bounded generators, and rotate post-RoPE queries and keys in learned low-rank head subspaces. Visual and non-visual positions use inverse frames. Values, the native mask, and the output projection retain their original paths. The final classifier reads caption-pooled and last-valid-position states. Layer indices are one-based.

the orthogonal complement of the learned subspace. A row vector is transformed as $xG = x + (xU)(R - I)U^\top$, without materializing a dense $d_h \times d_h$ matrix. QR and the solve both remain in the autograd graph.

C. Post-RoPE Relative Rotation and Local Invariance

The native layer first projects H^ℓ to queries, keys, and values, normalizes Q/K, and applies RoPE. Denote the resulting row vectors by q_i, k_i. Within each KV group, all four associated query heads share the same U and R. Suppressing layer and group indices, CMCR applies

$$F_i = \begin{cases} G^\top, & i \in \mathcal{V}, \\ G, & i \notin \mathcal{V}, \end{cases} \qquad q_i' = q_i F_i, \quad k_i' = k_i F_i. \tag{7}$$

Caption and template positions therefore use the non-visual frame, although only caption positions contribute to the text statistics. Values are unmodified. The original attention mask M and output projection are retained:

$$L'_{ij} = \frac{q_i F_i F_j^\top k_j^\top}{\sqrt{d_h}}, \qquad O' = \mathrm{softmax}(L' + M)V. \tag{8}$$

The native causal mask determines the permitted attention edges, while the full-caption controller conditions frames at all positions. CMCR therefore performs full-input discriminative prediction.

Local same-frame invariance. Fix a sample, layer input, and KV group. If $F_i = F_j$, orthogonality gives

$$q_i'(k_j')^\top = q_i F_i F_i^\top k_j^\top = q_i k_j^\top, \qquad \|q_i'\|_2 = \|q_i\|_2. \tag{9}$$

The analogous key-norm identity also holds. For a non-visual query and visual key, the relative frame is G^2, not G. If $\|A\|_2 \le \alpha$, the corresponding score satisfies

$$q_t'(k_v')^\top = q_t G^2 k_v^\top, \qquad |L'_{tv} - L_{tv}| \le \frac{4\alpha \|q_t\|_2 \|k_v\|_2}{\sqrt{d_h}}. \tag{10}$$

To see the bound, note that $R - I = 2(I - A)^{-1}A$ and $\|(I - A)^{-1}\|_2 \le 1$ for real skew-symmetric A. Orthogonal lifting preserves the nonzero singular values of $R - I$, giving $\|G - I\|_2 \le 2\alpha$; also $\|G + I\|_2 \le 2$. Apply Cauchy–Schwarz to $q_t(G - I)(G + I)k_v^\top$. The reverse unmasked direction uses $(G^\top)^2$.

The identities hold in exact arithmetic for fixed input vectors. Cross-frame score changes propagate through softmax normalization and subsequent hidden states; BF16 execution adds finite-precision effects. Figure 3 shows the preserved and modulated logit blocks together with the common-frame control.

D. Supervised Optimization and One-Pass Prediction

CMCR reads the final caption states and the final valid input position, which belongs to the existing assistant prefix. If s_* is that position, the classifier is

$$h_{\mathrm{read}} = \left[\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} H_i^L ; H_{s_*}^L \right], \qquad p(y \mid I, T) = \mathrm{softmax}(f_{\mathrm{cls}}(h_{\mathrm{read}})). \tag{11}$$

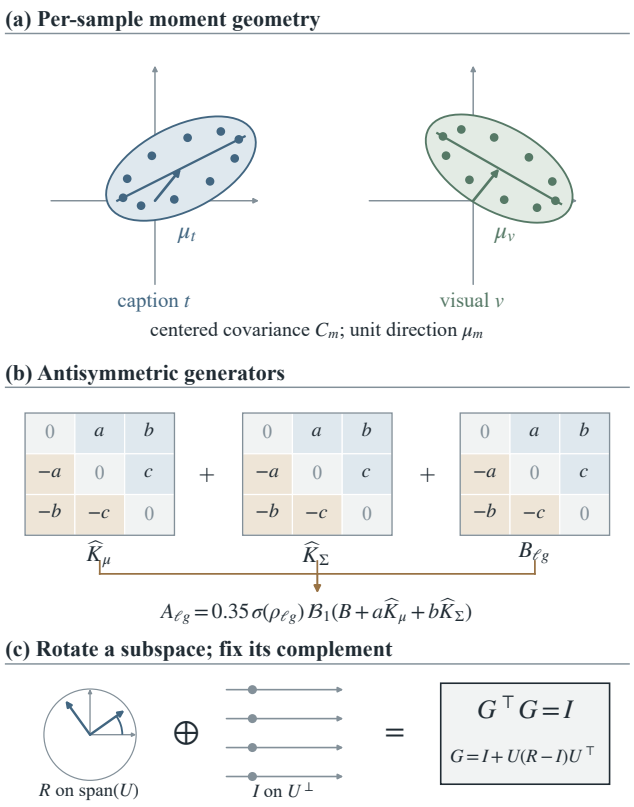


Fig. 2. Operator geometry in the statistical and head subspaces. First-order directions and second-order covariance operators produce distinct skew-symmetric terms. Bounded combination, Cayley rotation, and orthogonal lifting define a constrained change of frame.

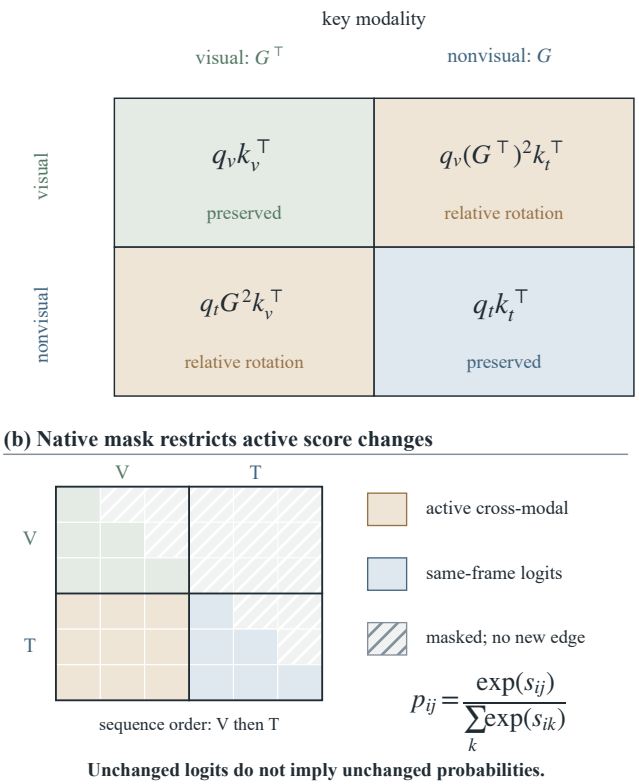


Fig. 3. Relative-frame attention and its controls. Same-frame logit blocks retain their fixed-input inner products; cross-frame blocks can change only where permitted by the native mask. Rotating both modalities by the same frame removes this direct relative effect in exact arithmetic.

The head consists of LayerNorm, an $8192 \to 256$ linear map, GELU, dropout 0.1, and a $256 \to 2$ linear map for the 8B model. The 4B model uses a 5120-dimensional input. The frozen backbone is adapted with rank-eight language-side LoRA; the geometry and classification head are trained jointly.

Let B be the minibatch size, $K = 3$ the number of insertion layers, and $H_k = 8$ the number of KV groups. The rotation-energy objective averages every entry of the small matrices:

$$\mathcal{L}_{\text{rot}} = \frac{1}{KBH_k r^2} \sum_{\ell,b,g} \|R_{\ell bg} - I_r\|_F^2. \tag{12}$$

This term encourages small transformations. The implemented objective is

$$\mathcal{L} = -\frac{1}{B} \sum_{b=1}^{B} \log p(y_b \mid I_b, T_b) + 10^{-3} \mathcal{L}_{\text{rot}}. \tag{13}$$

Inference selects $\hat{y} = \arg\max_c p(c \mid I, T)$ in one forward pass with KV caching disabled. Algorithms 1 and 2 specify training and fixed-checkpoint evaluation, respectively.

The added per-layer work comprises the statistical projection $O(Sdr)$, moment computation $O(Sr^2)$, groupwise small-matrix operations $O(H_k r^3)$, thin QR $O(H_k d_h r^2)$, and query/key projection and reconstruction $O((H_q + H_k) S d_h r)$. Native attention retains its original computational dependence on sequence length. The resource analysis measures the training-time and memory costs of these operations.

TABLE I
CLASS COMPOSITION OF THE FIXED, CLEANED MMSD2.0 PARTITIONS.

Partition	Non-sarcastic	Sarcastic	Total
Training	10,094	9,250	19,344
Validation	1,359	1,034	2,393
Test	1,372	1,037	2,409

IV. EXPERIMENTS

A. Dataset

We use MMSD2.0 [1], with the cleaned local partition summarized in Table I. Each item consists of its original image, original caption, and binary label; label one denotes sarcasm. Training and validation contain 19,344 and 2,393 items, respectively, and the test set contains 2,409 items. The same partition is used for every controlled full-data experiment. No generated captions, target boxes, pragmatic labels, or edited image–text pairs are required by CMCR.

The split audit checks cross-partition matches of IDs, lowercased whitespace-normalized captions, and exact image-file hashes, with zero intersections in each category. The training split contains 19,250 unique image hashes among 19,344 items. The reported intervals use test rows as the resampling unit. The test partition was previously inspected during development.

Algorithm 1 Supervised training with CMCR attention

Require: Image–caption training set; validation set; pre-trained backbone

Freeze base weights; add language LoRA and classification head
Initialize W_z, layer norms, controllers, P, U^0, and ρ
for each training minibatch **do**
  Build native inputs and caption/visual masks
  for each selected self-attention layer ℓ **do**
    Compute native Q/K/V, Q/K norm, then RoPE
    Compute Z, modality moments, and $\widehat{K}_\mu, \widehat{K}_\Sigma$
    Obtain group coefficients and bounded A by Eqs. (4)–(5)
    Solve $(I - A)R = I + A$; compute $U = \mathrm{qf}(U^0)$
    Rotate non-visual Q/K by G and visual Q/K by $G^\top$
    Call native attention with unchanged V and mask
  end for
  Pool final caption states and gather last valid state
  Compute class logits and loss in Eq. (13)
  Accumulate gradients; update trainable parameters
end for
Select the epoch with highest validation macro-F1

Algorithm 2 Single-pass classification and controlled evaluation

Require: Selected checkpoint, input (I, T), mode m

Set evaluation mode; disable KV caching
Form native input and the two modality masks
for each selected attention layer **do**
  if $m = \mathrm{bypass}$ **then**
    Retain native post-RoPE Q/K
  else
    Compute frame parameters from the current input
    For $m = \mathrm{no_comm}$, omit the explicit K_Σ term
    For $m = \mathrm{reverse}$, replace A by $-A$
    For $m = \mathrm{common}$, use G for all Q/K positions
    Otherwise apply the visual/non-visual inverse frames
  end if
  Continue native attention and later decoder layers
end for
Read h_{read}; return p and $\arg\max p$
Store ID, label, prediction, and both probabilities for evaluation

The principal measures are accuracy and macro-F1, with per-class precision, recall, and exact error counts. Probability quality is evaluated using negative log likelihood, the binary positive-class Brier score (mean squared error of $p(y = 1)$), and confidence ECE with ten equal-width bins on $[0, 1]$, following the confidence-calibration convention [31]. Empty bins contribute zero. Selective risk uses each model's own descending-confidence ranking, following the risk–coverage perspective of selective classification [32]. Shared confidence strata use the same Static-defined partition for both compared models. All headline results use the fixed argmax decision, equivalent to a positive-class threshold of 0.5.

TABLE II
FROZEN FULL-DATA TRAINING AND INFERENCE CONFIGURATION.

Setting	Value
Primary backbone	Qwen3-VL-8B-Thinking
Geometry rank / insertion layers	16 / 12, 24, 32
Q heads / KV groups / head width	32 / 8 / 128
LoRA rank / alpha / dropout	8 / 16 / 0.05
LoRA learning rate	5×10^{-5}
Geometry and head learning rate	2×10^{-4}
AdamW weight decay / gradient clip	0.01 / 1.0
Warm-up / scheduler	5% / cosine
Microbatch / accumulation	2 / 8
Epochs / optimizer steps	2 / 2,418
Weight seed / module seed	42 / 942
Image pixel range	100,352–200,704
Generator cap / energy weight	0.35 / 10^{-3}
Checkpoint selection	Validation macro-F1
Prediction / KV cache	Argmax / disabled

B. Baselines

We retain text-only, vision-only, conventional multimodal, and large vision–language reference systems. Tables IV and V summarize their historical benchmark results. The text and vision references include RoBERTa, TextCNN, and ViT; the discriminative multimodal group includes Multi-view CLIP, Att-BERT, CMGCN, HKE, HFM, and TFCD. The VLM group includes Qwen3.5, LLaVA variants, PaliGemma, InstructBLIP, and a retrieval-assisted LLaVA configuration. These reference results retain their original training recipes and F1 definitions.

The component variants are trained independently on the same Qwen3-VL backbone. *Static* uses only $B_{\ell g}$ in Eq. (5). *NoComm* retains the first-order term but omits the explicit covariance commutator. *CMCR* uses all three terms. The No-Comm descriptor retains both covariance diagonals, isolating the explicit commutator term in the generator. All groups allocate the same parameter tensors to control initialization; gradients follow each variant's active branches.

Fixed-checkpoint interventions probe the trained geometric pathway. Starting from the validation-selected CMCR checkpoint, bypass removes the rotations, no-commutator removes K_Σ, reverse negates A, and common-frame applies G to both modalities. These modes hold the trained weights fixed and use the same 128 test items selected with seed 3907.

C. Implementation Details

Our primary backbone is Qwen3-VL-8B-Thinking [33]. The configuration fixes a 4096-dimensional residual stream, 36 decoder blocks, 32 query heads, eight KV heads, and 128 dimensions per head. The accompanying 4B experiment uses the same insertion indices, rank, and data schedule with its native hidden width. Both use BF16 base weights and LoRA, with CMCR statistics, QR, linear solves, and the classifier computed in FP32. The visual encoder is frozen and excluded from LoRA targeting.

LoRA targets the language query, key, value, output, gate, up, and down projections. AdamW has separate learning rates for backbone adapters and custom parameters, weight decay 0.01, gradient clipping at 1.0, five-percent warm-up, and cosine decay. A microbatch of two with accumulation over

TABLE III
VALIDATION-SELECTED 8B CLASSIFIERS ON THE SAME 2,409 TEST ITEMS. SCORES ARE PERCENTAGES; EVERY RUN COMPLETES TWO EPOCHS.

Method	Acc.	Macro F1	Correct	Epoch
Static	87.05	86.95	2097	2
NoComm	86.92	86.81	2094	2
CMCR	87.21	87.12	2101	2

eight microsteps gives an effective batch of 16. Each full-data run performs 2,418 optimizer steps across two epochs. Non-reentrant gradient checkpointing limits memory use. Each variant runs on one GPU, with two GPU workers executing separate variants concurrently. Images retain the processor's native handling within the stated pixel bounds.

At each epoch boundary, the validation set determines the best macro-F1 checkpoint. The selected adapter and custom weights are reloaded before testing. All three 8B runs select epoch two. The three 4B runs also complete two epochs; Static selects epoch one, while NoComm and CMCR select epoch two.

Uncertainty for the CMCR–Static accuracy difference uses 10,000 paired test-row bootstrap draws with analysis seed 9172026. The exact two-sided McNemar test uses the discordant correctness counts for this comparison. Wilson intervals describe single-model accuracy or class recall. Probability-shift intervals use paired-row resampling of the fixed diagnostic subset. All intervals quantify test-sample variability conditional on the trained checkpoints. Aggregate accuracy and macro-F1 comparisons cover the full model and both component variants; classwise analyses use confusion counts. Paired, confidence-stratified, and probability-quality analyses use the aligned Static and CMCR predictions. Numerical analyses combine these counts, predictions, probabilities, and run logs.

V. RESULTS

A. Predictive Performance and Benchmark Context

Table III shows that CMCR achieves 87.21% accuracy and 87.12% macro-F1, leading both component variants at the 8B scale. The agreement between these two metrics indicates that the improvement in overall correctness is accompanied by stronger class-balanced performance. With the backbone, data split, and checkpoint-selection rule held constant, this comparison provides a focused assessment of generator composition within the same adaptation framework.

Table IV places CMCR against historical unimodal and discriminative multimodal systems. Its reported accuracy exceeds the strongest listed discriminative reference, Multi-view CLIP + TFCD, by 0.67 percentage points. The spread among the multimodal references suggests that access to both modalities does not by itself determine recognition quality. This makes comparison with stronger interaction-based systems particularly informative when assessing the proposed attention-level adaptation. Historical F1 scores retain their source definitions, so accuracy provides the common metric for this contextual comparison.

TABLE IV
HISTORICAL UNIMODAL AND DISCRIMINATIVE MULTIMODAL CONTEXT ON MMSD2.0. THE LAST ROW IS THE PRESENT CONTROLLED EVALUATION.

System	Input	Accuracy	F1
RoBERTa	T	79.66	76.21
TextCNN	T	71.61	69.52
ViT	V	72.02	69.72
Multi-view CLIP	TV	85.64	84.10
Att-BERT	TV	80.03	77.04
CMGCN	TV	79.83	76.90
HKE	TV	76.50	72.25
HFM	TV	70.57	66.88
Multi-view CLIP + TFCD	TV	86.54	84.31
CMCR (8B)	TV	87.21	87.12

Historical scores follow the MMSD2.0 and TFCD evaluations [1], [34]. Their F1 column retains the source definition; CMCR reports macro-F1. T: text; V: image.

Table V extends the comparison to large vision–language systems. CMCR attains the highest reported accuracy and macro-F1 among the listed models, exceeding the retrieval-assisted LLaVA reference by 0.78 percentage points on both measures while retaining a single-pass classifier over the original image–caption pair. Its classwise profile is distinguished by high sarcastic recall and a smaller gap between the two class recalls than the references with available classwise results. This more even recall profile is useful for a task in which missed sarcasm can be obscured by strong recognition of the more frequent non-sarcastic class. The classwise results therefore clarify the source of the favorable macro-F1 ranking. These cross-system results provide benchmark context under the original evaluation recipes; Table III examines the geometric components under a shared training protocol.

B. First- and Second-Order Contributions across Scales

Table VI shows that CMCR outperforms NoComm in both accuracy and macro-F1 at both backbone sizes. Since NoComm retains first-order conditioning and covariance diagonals in its controller, this consistent ordering supports the inclusion of the explicit covariance commutator in the generator. The distinction is therefore between supplying covariance summaries to the controller and explicitly using their noncommuting interaction to orient the rotation. The wider margins at 8B suggest that the combined moment-based geometry benefits more from the higher-capacity backbone in the evaluated configurations.

Static remains slightly ahead at 4B, whereas CMCR leads at 8B and achieves the largest cross-scale accuracy gain (1.08 percentage points). This change in ranking suggests that the benefit of sample conditioning depends on its interaction with backbone capacity. The most consistent component-level finding is the advantage of the complete generator over the no-commutator variant; the strongest overall result emerges when that generator is paired with the larger backbone.

C. Paired Decision Changes Relative to Static Frames

Table VII shows a favorable balance of corrected and newly incorrect decisions for CMCR, while the two classifiers retain

TABLE V
LARGE VISION–LANGUAGE REFERENCE SYSTEMS AND THE NEW CMCR CLASSIFIER. SCORES ARE PERCENTAGES.

	Overall		Sarcastic class		Non-sarcastic class	
System	Macro-F1	Accuracy	Recall	Precision	Recall	Precision
Qwen3.5-4B [35]	79.37	80.38	67.70	83.60	89.97	78.65
Qwen3.5-9B [35]	82.43	83.20	72.32	86.44	91.42	81.38
LLaVA-OneVision-7B [36]	82.28	83.41	67.55	91.73	95.40	79.55
LLaVA-1.5-7B [37]	80.49	81.76	65.33	89.45	94.18	78.23
PaliGemma-3B [38]	79.74	80.88	66.39	86.00	91.83	78.33
InstructBLIP-7B [17]	78.36	79.14	69.87	79.22	86.15	79.09
LLaVA-1.5-7B + retrieval [8]	86.34	86.43	–	–	–	–
CMCR / Qwen3-VL-8B-Thinking	87.12	87.21	91.51	81.18	83.97	92.90

Reference scores summarize previously recorded or published evaluations. A dash denotes an unreported classwise value. Table III reports the CMCR, Static, and NoComm component comparison.

TABLE VI
ACCURACY AND MACRO-F1 ACROSS BACKBONE SCALES. VALUES ARE PERCENTAGES; ACCURACY GAINS ARE CALCULATED BEFORE ROUNDING.

	4B		8B		Acc. gain
Method	Acc.	Macro-F1	Acc.	Macro-F1	(pp)
Static	86.30	86.17	87.05	86.95	+0.75
NoComm	85.97	85.87	86.92	86.81	+0.95
CMCR	86.14	86.03	87.21	87.12	+1.08

TABLE VII
PAIRED 8B DECISION CHANGES BETWEEN CMCR AND STATIC. GAIN/LOSS COUNTS DENOTE ONLY CMCR CORRECT / ONLY STATIC CORRECT; p_{exact} IS THE EXACT TWO-SIDED MCNEMAR PROBABILITY. THE ACCURACY DIFFERENCE USES A 10,000-DRAW PAIRED-BOOTSTRAP 95% INTERVAL.

Control	Gain/loss	Δ (pp)	95% CI	p_{exact}
Static	14/10	+0.17	[-0.25,+0.58]	0.541

TABLE VIII
FIXED-CHECKPOINT INTERVENTIONS ON 128 SHARED TEST ITEMS. PROBABILITY DIFFERENCES USE THE INTACT MODEL AS REFERENCE; FLIPS COUNTS CHANGED PREDICTED LABELS.

Intervention	Acc.	Flips	Mean $\lvert\Delta p\rvert$	Max $\lvert\Delta p\rvert$
Intact	85.94	0	0.00e+00	0.00e+00
Bypass	85.94	0	1.50e-03	2.36e-02
No commutator	85.94	0	1.16e-03	1.40e-02
Reverse	85.94	0	1.67e-03	2.05e-02
Common frame	85.94	0	1.37e-03	2.14e-02

TABLE IX
CLASSWISE BEHAVIOR OF CMCR AND ITS STATIC-FRAME VARIANT AT THE 8B SCALE. EACH CLASS USES THE FIXED TEST SUPPORT IN TABLE I.

Method	Class	Precision	Recall	F1
Static	Non-sarc.	92.60	83.97	88.07
Static	Sarc.	81.12	91.13	85.83
CMCR	Non-sarc.	92.90	83.97	88.21
CMCR	Sarc.	81.18	91.51	86.04

the same label on 99.00% of the test set. The gain therefore comes from a localized adjustment within an otherwise stable prediction pattern. The classwise decomposition further shows that the favorable correction balance comes from sarcastic examples, whereas corrections and losses offset within the non-sarcastic class. This is consistent with a targeted recall improvement at the shared default threshold. The paired interval includes zero, so the observed advantage is interpreted as a modest improvement for the evaluated checkpoints, alongside the cross-scale and classwise evidence.

D. Relative-Frame Interventions and Numerical Controls

Table VIII separates probability sensitivity from changes in the selected class. All interventions retain the intact model's labels on the 128-item diagnostic subset, while mean absolute probability shifts remain on the order of 10^{-3}. The trained classifier thus exhibits decision stability under the tested frame interventions at the fixed threshold.

The common-frame control supplies the numerical context for these shifts. Common-frame and bypass are equivalent for fixed-input logits in exact arithmetic, yet their mean absolute probability difference in the executed network is 1.753×10^{-3}, comparable to the mode-versus-intact shifts. The observed probability responses should therefore be interpreted against this finite-precision reference. These probes document the stability of the learned decisions on the evaluated subset, while the independently trained variants assess the contribution of each component after optimization.

E. Classwise Sensitivity, Confidence, and Decision Thresholds

Table IX identifies the main predictive advantage of CMCR: improved sarcastic recognition at unchanged non-sarcastic recall. Both classwise F1 scores increase relative to Static, indicating a favorable precision–recall adjustment across the two classes. Recovering missed sarcastic instances also improves the precision of the predicted non-sarcastic class, whose recall remains unchanged. This explains how better recognition of sarcasm benefits both classwise F1 scores under the same decision threshold, while preserving the aggregate false-positive count.

Table X localizes the decision changes to the least-confident Static predictions. Using the same Static-defined groups holds the evaluated examples fixed within each confidence range, allowing the local accuracy improvement to be traced to changed classifications on those examples. Accuracy in the $[0.5, 0.7)$ stratum rises by 2.45 percentage points, while the

TABLE X
SHARED CONFIDENCE STRATA FIXED BY STATIC PROBABILITIES. ACCURACY IS REPORTED IN PERCENT; GAIN/LOSS DENOTES ONLY CMCR CORRECT / ONLY STATIC CORRECT WITHIN EACH STRATUM.

Static confidence	n	Static	CMCR	Gain/loss
$[0.5, 0.7)$	163	53.37	55.83	14/10
$[0.7, 0.9)$	249	67.07	67.07	0/0
$[0.9, 1.0]$	1997	92.29	92.29	0/0

TABLE XI
PROBABILITY QUALITY OF STATIC AND CMCR. ECE IS A PERCENTAGE; BRIER USES THE POSITIVE-CLASS SQUARED-ERROR CONVENTION. HC ERRORS HAVE MAXIMUM CLASS PROBABILITY AT LEAST 0.9.

Method	NLL	Brier	ECE	HC errors
Static	0.4983	0.1028	7.48	154
CMCR	0.4972	0.1028	7.39	159

more confident strata retain identical labels. Although this stratum contains only 6.77% of the test set, it accounts for the full net improvement. The larger gain within this small group suggests that the predictive benefit is concentrated on borderline inputs, providing a more informative account of the conditional adaptation than the aggregate accuracy difference alone.

Table XI shows slightly lower NLL and ECE for CMCR, with the same Brier score at the reported precision. The classification improvement is therefore accompanied by broadly comparable probability quality and small gains in the aggregate probability metrics. The high-confidence error count is marginally higher, so these gains are best understood as a modest refinement rather than a uniform improvement across confidence regions.

Table XII shows that the precision–recall balance can be adjusted at inference without retraining. Across thresholds 0.5–0.8, accuracy and macro-F1 remain within narrow bands as the decision rule shifts toward higher precision. This flexibility is useful when applications place different costs on missed sarcasm and false alarms. The sweep characterizes the fixed checkpoint; headline comparisons retain $\tau = 0.5$, and an application-specific threshold should be selected on validation data.

F. Geometric Activity and Computational Footprint

The finite-tensor operator audit verifies the intended local identities to floating-point tolerance, including same-frame logit preservation, common-frame cancellation, and the cross-frame perturbation bound. Table XIII complements this audit with the recorded optimization diagnostics: both conditional gate coefficients are nonzero in the final microbatch, and the full model exhibits greater rotation magnitude than either component variant. These measurements document the realized geometric activity within the bounded low-rank construction.

Table XIV shows that this additional activity has a small memory footprint. All variants retain the same allocated trainable-parameter count, including tensors inactive in some controls, and CMCR adds only 0.020 GiB of peak memory relative to Static, approximately 0.12%. The unchanged recorded

TABLE XII
DECISION-THRESHOLD RESPONSE OF THE FIXED CMCR CHECKPOINT.

τ	Acc.	Macro F1	Precision	Recall
0.1	83.44	83.44	73.22	97.01
0.2	85.68	85.66	76.91	95.37
0.3	86.51	86.47	78.62	94.31
0.4	86.92	86.86	79.98	92.86
0.5	87.21	87.12	81.18	91.51
0.6	87.17	87.04	82.16	89.68
0.7	87.34	87.17	83.46	88.04
0.8	87.42	87.18	85.22	85.63
0.9	86.26	85.85	86.69	80.42

TABLE XIII
LAST RECORDED 8B OPTIMIZATION DIAGNOSTICS. ENERGY AVERAGES SQUARED ENTRIES OF $R - I$ OVER THE LAST LOGGING WINDOW; THE FROBENIUS QUANTITY AND ACTIVE GATE COEFFICIENTS SUMMARIZE ITS FINAL MICROBATCH. DASHES DENOTE INACTIVE TERMS.

Method	Energy	$\|R - I\|_F$	$\bar{a}_\mu$	$\bar{a}_C$
Static	3.75e-06	0.0306	–	–
NoComm	6.06e-05	0.1248	0.6262	–
CMCR	1.15e-04	0.1734	0.6227	0.6273

peak memory relative to NoComm further indicates that the second-order term fits within a similar memory budget, a useful property when adapting larger backbones under constrained GPU memory. Computation time is the principal measured increment: the full model's recorded optimization window is 16.5% longer than Static's. This window includes validation and checkpoint operations on a shared host and therefore describes the recorded training workload. At inference, CMCR retains single-pass classification.

G. Prediction Stability and Error Structure

Table XV shows substantial overlap in both correct predictions and residual errors between CMCR and Static. The configuration-specific outcomes favor CMCR, while most established decisions are retained. Read alongside the shared confidence strata in Table X, this overlap supports an interpretation of conditional geometry as a focused refinement of uncertain decisions. Its clearest observed benefit is improved sarcastic recognition within a largely preserved prediction structure.

VI. DISCUSSION

A. Relative Attention Geometry and Information Flow

CMCR constrains attention interactions through modality-relative orthogonal frames. At a fixed insertion input, a shared orthogonal transformation preserves same-frame Q/K inner products, while inverse visual and non-visual frames allow cross-frame scores to change. Value vectors and the original mask follow the backbone's native computation.

Cross-frame logit changes redistribute normalized attention weights and propagate through subsequent layers. The controller uses the complete caption to construct sample-conditioned frames for full-input classification.

Moment statistics describe the distribution of hidden vectors in the learned subspace. Modality means capture first-order

TABLE XIV
RECORDED 8B OPTIMIZATION-WINDOW COST AND ALLOCATED TRAINABLE PARAMETERS. TIMING INCLUDES VALIDATION AND CHECKPOINT OPERATIONS.

Method	Time (h)	Peak (GiB)	Alloc. (M)
Static	3.51	17.146	24.105
NoComm	4.40	17.166	24.105
CMCR	4.09	17.166	24.105

TABLE XV
PREDICTION-CORRECTNESS OVERLAP BETWEEN CMCR AND STATIC ON THE 2,409-ITEM TEST SET.

Prediction outcome	Cases	Test share (%)
Both correct	2087	86.63
Only CMCR correct	14	0.58
Only Static correct	10	0.42
Both incorrect	298	12.37

directions, and covariance commutators encode the noncommuting part of the second-order interaction. These quantities provide complementary inputs to the sample-conditioned rotation.

Figure 4 connects backbone-scale accuracy to classwise sensitivity. Its first panel compares CMCR with its two component variants at 4B and 8B using Wilson intervals computed from correct counts. The second panel compares the full and static-frame models within the two 8B label classes. It shows increased sarcastic recall alongside unchanged non-sarcastic recall, linking the overall gain to class-level recognition behavior.

B. Complementary Component and Numerical Controls

Independently trained controls assess the contribution of each component after optimization, while fixed-checkpoint interventions probe the response of a learned model to that component. NoComm learns through its static term, first-order signal, controller, and backbone adapters; the no-commutator intervention changes only the corresponding term in the selected CMCR checkpoint. Together, these analyses characterize component learning and local sensitivity.

The common-frame control adds a numerical reference to the component analysis. Its exact-arithmetic direct effect cancels, so its comparison with bypass characterizes finite-precision differences and their propagation. Probability shifts and retained labels then describe distinct aspects of the trained classifier's response. Figure 5 displays these intervention shifts beside their common-frame reference, separating distributional sensitivity from final-decision behavior.

The cross-scale analysis shows that the full CMCR configuration has the largest accuracy gain when moving from 4B to 8B. The component comparison associates the 8B net improvement over Static with low-confidence decisions and increased sarcastic recall. The shared-error partition identifies which outputs remain stable across these configurations. Together, these analyses connect the generator's composition with the distribution of its observed prediction changes.

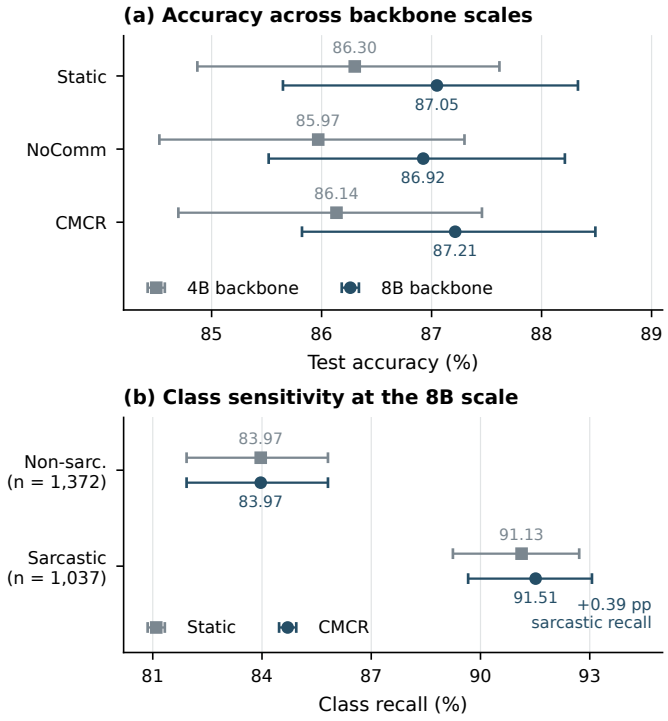


Fig. 4. Backbone-scale accuracy and classwise sensitivity. (a) Accuracy of CMCR and its two component variants at 4B and 8B, with Wilson 95% intervals. (b) Recall of Static and CMCR for the two 8B classes, with Wilson 95% intervals using the corresponding class supports.

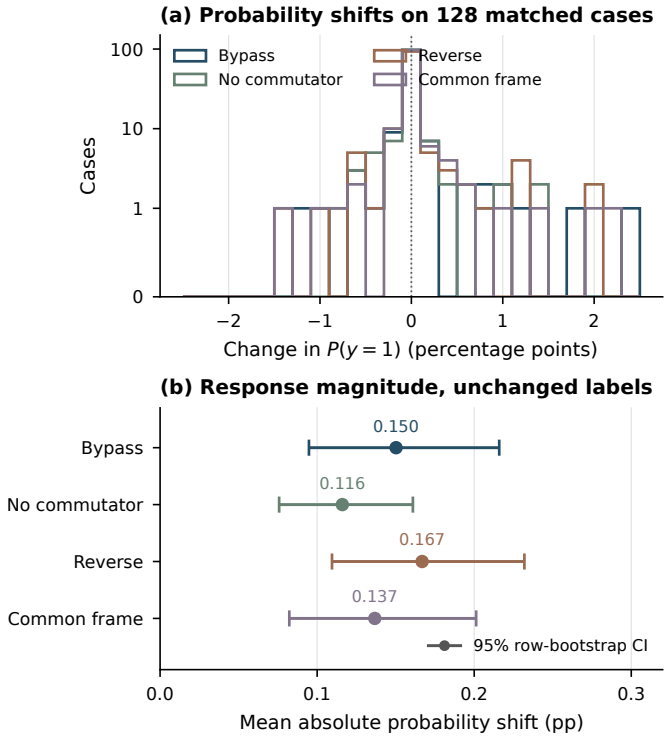


Fig. 5. Intervention sensitivity and its numerical context. (a) Signed positive-class probability shifts relative to intact CMCR on 128 shared items, with a symmetric-log count axis. (b) Mean absolute shifts with paired-bootstrap 95% intervals. All interventions retain the original predicted labels; the common-frame control provides a finite-precision reference.

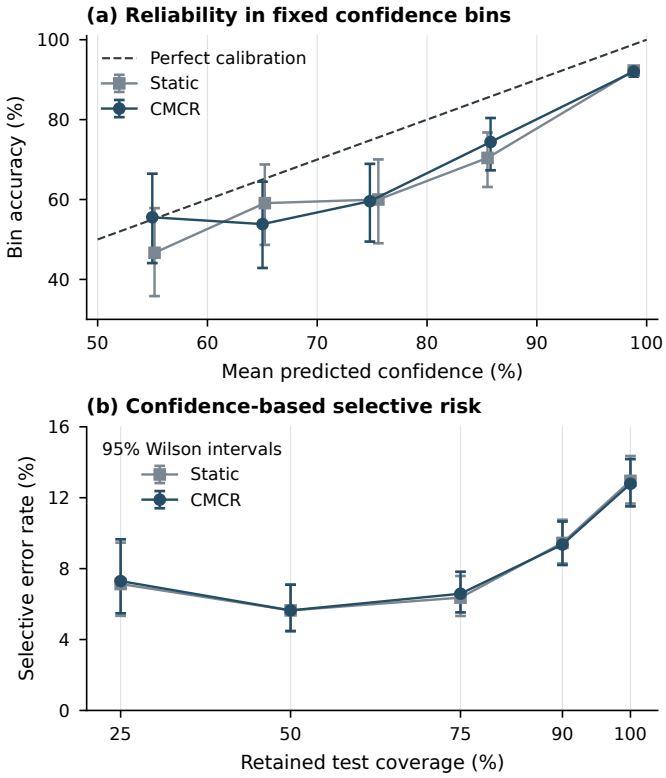


Fig. 6. Confidence reliability and selective prediction for Static and CMCR. (a) Reliability of saved test probabilities, with bin-wise Wilson intervals and an identity reference. (b) Error rates after retaining each model's highest-confidence items at fixed coverage, with Wilson 95% intervals.

Probability quality provides another view of the classifier. Figure 6 places the Static and CMCR reliability curves beside their selective-risk profiles, describing calibration and error rates under confidence-based retention. Each model defines its own ranking for selective prediction, whereas the confidence strata in Table X use a shared Static-defined partition.

The paired Static comparison in Table VII connects the accuracy gain to corrected errors and newly incorrect predictions. Together with the overlap counts in Table XV, it resolves the prediction-level effect of conditional rotation into stable, corrected, and newly mistaken outputs.

C. Extensions of Geometric Adaptation

Further evaluation can cover multiple training seeds and external benchmarks, with rank, insertion depth, generator cap, and energy weight selected on validation data. Parameter-active matched non-orthogonal controllers can isolate the role of orthogonality within conditional adaptation. Image-cluster resampling and near-duplicate auditing can extend the current item-level analysis.

Precision-controlled layer-local probes can evaluate Eq. (9) numerically and trace cross-frame changes through subsequent layers to the classifier. Region-annotated image–text examples and targeted modality interventions can relate these changes to task-relevant visual and linguistic cues.

Basis factorizations, grouped projection reuse, and removal of inactive diagnostic computations are potential routes to lower overhead. Deployment-oriented evaluation can measure inference latency under controlled batch size, image resolution, and sequence length.

VII. Limitations

CMCR uses complete image–text inputs to construct sample-conditioned attention rotations. Incremental generation would require sequential conditioning and key–value cache compatibility. The current fixed subspace ranks and insertion layers also motivate adaptive configurations for different architectures and input structures. The geometric guarantee concerns fixed-input attention logits in exact arithmetic. Region-level analysis and targeted modality interventions can further characterize the visual and linguistic interactions associated with the learned rotations.

VIII. Conclusion

We introduced Cross-Modal Commutator Rotation, a sample-conditioned attention adaptation mechanism for multimodal sarcasm detection. First-order modality directions, covariance commutators, and a learned static term define bounded skew-symmetric generators. Cayley transforms and learned orthogonal subspaces turn them into inverse visual and non-visual Q/K frames after positional encoding. The operator preserves fixed-input same-frame logits in exact arithmetic while permitting constrained cross-frame changes. Classification uses existing caption and assistant-prefix states in a single forward pass.

On the cleaned MMSD2.0 split, the 8B CMCR classifier achieves 87.21% accuracy and 87.12% macro-F1. The complete formulation leads its static-frame and no-commutator variants at 8B, while exceeding the no-commutator variant at both backbone scales. Classwise and paired analyses connect the improvement to increased sarcastic recall, gains in both classwise F1 scores, and decision changes concentrated in the least-confident stratum. Cross-scale evaluation, common-frame interventions, and confidence and resource measurements characterize the model's predictive behavior and geometric computations. CMCR provides a unified framework for multimodal sarcasm classification through sample-conditioned attention geometry.

References

[1] L. Qin, S. Huang, Q. Chen *et al.*, "MMSD2.0: Towards a Reliable Multi-Modal Sarcasm Detection System," in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 10 834–10 845. [Online]. Available: https://aclanthology.org/2023.findings-acl.689/

[2] Y. Cai, H. Cai, and X. Wan, "Multi-Modal Sarcasm Detection in Twitter with Hierarchical Fusion Model," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 2506–2515. [Online]. Available: https://aclanthology.org/P19-1239/

[3] S. Farabi, T. Ranasinghe, D. Kanojia, Y. Kong, and M. Zampieri, "A Survey of Multimodal Sarcasm Detection," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, 2024, pp. 8020–8028. [Online]. Available: https://www.ijcai.org/proceedings/2024/887

[4] C. Wen, G. Jia, and J. Yang, "DIP: Dual Incongruity Perceiving Network for Sarcasm Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 2540–2550. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2023/html/Wen_DIP_Dual_Incongruity_Perceiving_Network_for_Sarcasm_Detection_CVPR_2023_paper.html

[5] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139, 2021, pp. 8748–8763. [Online]. Available: https://proceedings.mlr.press/v139/radford21a.html

[6] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., vol. 34. Curran Associates, Inc., 2021, pp. 9694–9705. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/505259756244493872b7709a8a01b536-Paper.pdf

[7] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proceedings of the 40th International Conference on Machine Learning*, vol. 202, 2023, pp. 19 730–19 742. [Online]. Available: https://proceedings.mlr.press/v202/li23q.html

[8] B. Tang, B. Lin, H. Yan, and S. Li, "Leveraging generative large language models with visual instruction and demonstration retrieval for multimodal sarcasm detection," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 1732–1742. [Online]. Available: https://aclanthology.org/2024.naacl-long.97/

[9] Z. Chen, H. Lin, Z. Luo, M. Cheng, J. Ma, and G. Chen, "CofiPara: A coarse-to-fine paradigm for multimodal sarcasm target identification with large multimodal models," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 9663–9687. [Online]. Available: https://aclanthology.org/2024.acl-long.522/

[10] Z. Qiu, W. Liu, H. Feng *et al.*, "Controlling text-to-image diffusion by orthogonal finetuning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 79 320–79 362. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/faacb7a4827b4d51e201666b93ab5fa7-Abstract-Conference.html

[11] Z. Wu, A. Arora, Z. Wang *et al.*, "ReFT: Representation finetuning for language models," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/75008a0fba53bf13b0bb3b7bff986e0e-Abstract-Conference.html

[12] H. Pan, Z. Lin, P. Fu, Y. Qi, and W. Wang, "Modeling Intra and Inter-modality Incongruity for Multi-Modal Sarcasm Detection," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1383–1392. [Online]. Available: https://aclanthology.org/2020.findings-emnlp.124/

[13] H. Liu, W. Wang, and H. Li, "Towards Multi-Modal Sarcasm Detection via Hierarchical Congruity Modeling with Knowledge Enhancement," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 4995–5006. [Online]. Available: https://aclanthology.org/2022.emnlp-main.333/

[14] Y. Tian, N. Xu, R. Zhang, and W. Mao, "Dynamic Routing Transformer Network for Multimodal Sarcasm Detection," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 2468–2480. [Online]. Available: https://aclanthology.org/2023.acl-long.139/

[15] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019. [Online]. Available: https://doi.org/10.1109/TPAMI.2018.2798607

[16] P. Xu, X. Zhu, and D. A. Clifton, "Multimodal learning with transformers: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 10, pp. 12 113–12 132, 2023. [Online]. Available: https://doi.org/10.1109/TPAMI.2023.3275156

[17] W. Dai, J. Li, D. Li *et al.*, "InstructBLIP: Towards General-Purpose Vision-Language Models with Instruction Tuning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 49 250–49 267. [Online]. Available: https://papers.neurips.cc/paper_files/paper/2023/hash/9a6a435e75419a836fe47ab6793623e6-Abstract-Conference.html

[18] A. J. Wang, P. Zhou, M. Z. Shou, and S. Yan, "Enhancing visual grounding in vision-language pre-training with position-guided text prompts," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3406–3421, 2024. [Online]. Available: https://doi.org/10.1109/TPAMI.2023.3343736

[19] E. J. Hu, Y. Shen, P. Wallis *et al.*, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022. [Online]. Available: https://openreview.net/forum?id=nZeVKeeFYf9

[20] S.-Y. Liu, C.-Y. Wang, H. Yin *et al.*, "DoRA: Weight-decomposed low-rank adaptation," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 32 100–32 121. [Online]. Available: https://proceedings.mlr.press/v235/liu24bn.html

[21] D. Kopiczko, T. Blankevoort, and Y. Asano, "VeRA: Vector-based random matrix adaptation," in *International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/1b53ad08de383a049e9668a9d0b6a053-Abstract-Conference.html

[22] W. Liu, Z. Qiu, Y. Feng *et al.*, "Parameter-efficient orthogonal finetuning via butterfly factorization," in *International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/pdf/717cca7ed2cd7fd840370c3e76a702396744c7ea.pdf

[23] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu, "RoFormer: Enhanced transformer with rotary position embedding," *Neurocomputing*, vol. 568, p. 127063, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231223011864

[24] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebron, and S. Sanghai, "GQA: Training generalized multi-query transformer models from multi-head checkpoints," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 4895–4901. [Online]. Available: https://aclanthology.org/2023.emnlp-main.298/

[25] Y. Li, Y. Du, K. Zhou, J. Wang, W. X. Zhao, and J.-R. Wen, "Evaluating Object Hallucination in Large Vision-Language Models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 292–305. [Online]. Available: https://aclanthology.org/2023.emnlp-main.20/

[26] W. Yu, Z. Yang, L. Li *et al.*, "Mm-vet: Evaluating large multimodal models for integrated capabilities," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 57 730–57 754. [Online]. Available: https://proceedings.mlr.press/v235/yu24o.html

[27] S. Leng, H. Zhang, G. Chen *et al.*, "Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 13 872–13 882. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Leng_Mitigating_Object_Hallucinations_in_Large_Vision-Language_Models_through_Visual_Contrastive_CVPR_2024_paper.html

[28] Q. Huang, X. Dong, P. Zhang *et al.*, "Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 13 418–13 427. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Huang_OPERA_Alleviating_Hallucination_in_Multi-Modal_Large_Language_Models_via_Over-Trust_CVPR_2024_paper.html

[29] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," in *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., vol. 36. Curran Associates, Inc., 2023, pp. 53 728–53 741. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/file/a85b405ed65c6477a4fe8302b5e06ce7-Paper-Conference.pdf

[30] S. Jain and B. C. Wallace, "Attention is not explanation," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 3543–3556. [Online]. Available: https://aclanthology.org/N19-1357/

[31] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, vol. 70, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[32] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

[33] S. Bai, Y. Cai, R. Chen *et al.*, "Qwen3-VL Technical Report," *arXiv preprint arXiv:2511.21631*, 2025. [Online]. Available: https://arxiv.org/abs/2511.21631

[34] Z. Zhu, X. Zhuang, Y. Zhang *et al.*, "TFCD: Towards multi-modal sarcasm detection via training-free counterfactual debiasing," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, 2024, pp. 6687–6695. [Online]. Available: https://www.ijcai.org/proceedings/2024/739

[35] Qwen Team, "Qwen3.5: Towards native multimodal agents," February 2026. [Online]. Available: https://qwen.ai/blog?id=qwen3.5
[36] B. Li, Y. Zhang, D. Guo *et al.*, "LLaVA-OneVision: Easy Visual Task Transfer," *Transactions on Machine Learning Research*, 2025. [Online]. Available: https://openreview.net/forum?id=zKv8qULV6n
[37] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved Baselines with Visual Instruction Tuning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 296–26 306. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Liu_Improved_Baselines_with_Visual_Instruction_Tuning_CVPR_2024_paper.html
[38] L. Beyer, A. Steiner, A. S. Pinto *et al.*, "PaliGemma: A versatile 3B VLM for transfer," *arXiv preprint arXiv:2407.07726*, 2024. [Online]. Available: https://arxiv.org/abs/2407.07726



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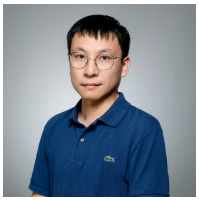
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